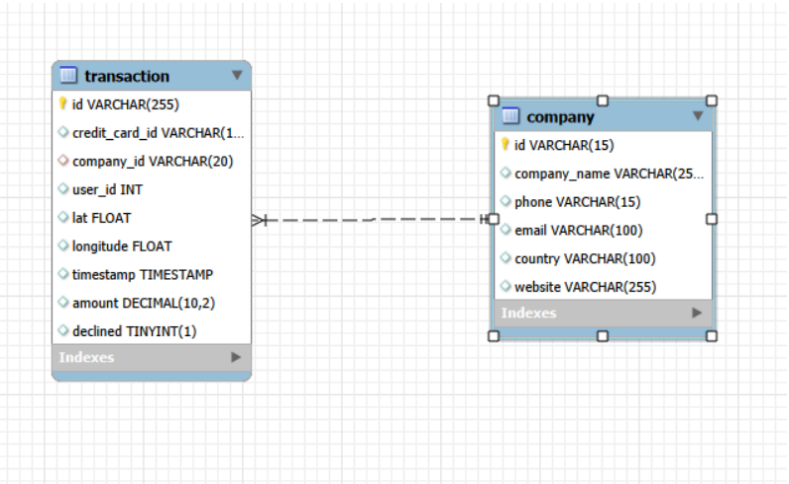
Nivell 1

Exercici 1

A partir dels documents adjunts (estructura_dades i dades_introduir), importa les dues taules. Mostra les característiques principals de l'esquema creat i explica les diferents taules i variables que existeixen. Assegura't d'incloure un diagrama que il·lustri la relació entre les diferents taules i variables.



La base de datos transactions se compone de 2 tablas: transaction y company, que tienen una relación de n a 1, ya que la primera (transacciones) funciona como tabla de hechos registrando transacciones realizadas por distintas empresas que vienen descritas en la tabla de dimensiones company.

Su relación se establece en transaction.company_id y company.id

Tabla transactions

Field	Type	Null	Key
id	varchar(255)	NO	PRI
credit_card_id	varchar(15)	YES	
company_id	varchar(20)	YES	MUL
user_id	int	YES	
lat	float	YES	
longitude	float	YES	
timestamp	timestamp	YES	
amount	decimal(10,2)	YES	
declined	tinyint(1)	YES	

La tabla se describe con las siguientes columnas y tipos de variables:

- id: asigna un identificador único a la transacción, es la primary key (PK), no puede tener valores nulos y es de tipo varchar(255), lo que significa que puede albergar valores de hasta 255 caracteres.
- credit_card_id: contiene variables tipo varchar(15) lo que le permite almacenar hasta 15 caracteres, este campo sí permite valores nulos.

- company_id: este campo funciona como foreign key (FK) en relación con la tabla company, por lo que admite que se repita múltiples veces dentro de él (MUL), es de tipo varchar(20) y admite nulos.
- user_id: es una variable de tipo numérico entero y admite nulos.
- lat: contiene el valor de la latitud y es de tipo numérico float (decimal), admite nulos.
- longitude: al igual que la columna anterior pero almacena la longitud geográfica, también es de tipo float y admite nulos.
- timestamp: almacena en formato timestamp (yyyy-mm-dd hh:mm:ss) los datos de fecha y hora de la transacción. Admite nulos.
- amount: almacena el monto de la transacción, es tipo decimal(10,2) lo que significa que contará con hasta 10 dígitos antes del punto decimal y 2 después de este. Admite nulos.
- declined: de tipo tinyint(1), un valor numérico entero que sólo admite un dígito (0 o 1) por lo que funciona como boleano equivalente a valores True / False.

Tabla company

	Field	Type	Null	Key
▶	id	varchar(15)	NO	PRI
	company_name	varchar(255)	YES	
	phone	varchar(15)	YES	
	email	varchar(100)	YES	
	country	varchar(100)	YES	
	website	varchar(255)	YES	

La tabla se describe con las siguientes columnas y tipos de variables:

- id: es la primary key (PK) de la tabla, asigna un identificador a cada empresa, es de tipo varchar(15) por lo que admite hasta 15 caracteres, no admite nulos.
- company_name: almacena el nombre de la empresa, es de tipo varchar(255) admite hasta 255 caracteres, admite nulos.
- phone: contiene el número de teléfono de la empresa, es de tipo varchar(15), para hasta 15 caracteres, admite nulos.
- email: almacena el email de la empresa, es de tipo varchar(100), para hasta 100 caracteres, admite nulos.
- country: contiene el país de la empresa, es de tipo varchar(100) al igual que la columna anterior, admite nulos.
- website: almacena la dirección web de la empresa, es de tipo varchar(255) para hasta 255 caracteres, admite nulos.

Exercici 2

Utilitzant JOIN realitzaràs les següents consultes:

Llistat dels països que estan generant vendes.

```
SELECT DISTINCT c.country
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id;
```

country
Germany
Australia
United States
New Zealand
Norway
United Kingdom
Italy
Belgium
Sweden
Ireland
China
Canada
France
Netherlands
Spain

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
4	22:28:02	SELECT DISTINCT c.country FROM company c JOIN transaction t ON c.id = t.company_id LIMIT 0, 2000	15 row(s) returned	0.000 sec / 0.000 sec

Des de quants països es generen les vendes.

```
SELECT COUNT(DISTINCT c.country) AS num_company_country
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id;
```

num_company_country
15

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Or

Output

Action Output

#	Time	Action	Message	Duration / Fetch
5	22:28:02	SELECT COUNT(DISTINCT c.country) AS num_company_country FROM company c JOIN transaction t ON c.id = t.company_id LIMIT 0, 2000	1 row(s) returned	0.109 sec / 0.000 sec

Identifica la companyia amb la mitjana més gran de vendes.

```
SELECT c.company_name, ROUND(AVG(amount),2) AS media_ventas
  FROM company c
  JOIN transaction t
    ON c.id = t.company_id
 GROUP BY c.id
 ORDER BY media_ventas DESC
 LIMIT 1;
```

company_name	media_ventas
Ac Fermentum Incorporated	284.87

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Or

Output

Action Output

#	Time	Action	Message	Duration / Fetch
6	22:28:02	SELECT c.company_name, ROUND(AVG(amount),2) AS media_ventas FROM company c JOIN transaction t ON c.id = t.company_id GROUP B...	1 row(s) returned	0.312 sec / 0.000 sec

Exercici 3

Utilitzant només subconsultes (sense utilitzar JOIN):

Mostra totes les transaccions realitzades per empreses d'Alemanya.

```
38 •  SELECT t.id,
39            t.company_id,
40            (SELECT c.company_name
41               FROM company c
42              WHERE c.id=t.company_id) AS company
43      FROM transaction t
44     WHERE EXISTS (
45                    SELECT 1
46                      FROM company c
47                     WHERE c.id = t.company_id
48                           AND country = 'Germany'
49                   );
50
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

id	company_id	company
00138D3B-206D-4C03-94B7-63A2676EB9B4	b-2222	Ac Fermentum Incorporated
0013C1B6-3B84-4D6C-8154-E2B3FEBCA8E9	b-2222	Ac Fermentum Incorporated
00201A11-2E62-44C4-941D-198FC8DB77F0	b-2222	Ac Fermentum Incorporated
00235618-0A5C-4D49-9DCB-B3A9405D8923	b-2222	Ac Fermentum Incorporated
005A5A7B-1F1A-4B6C-9B15-1625A78C9C38	b-2222	Ac Fermentum Incorporated

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read On

Output

Action Output

#	Time	Action	Message	Duration / Fetch
7	22:28:02	SELECT t.id, t.company_id, (SELECT c.company_name FROM company c WHERE c.id=t.company_id) AS company FROM transacti...	2000 row(s) returned	0.000 sec / 0.015 sec

Llista les empreses que han realitzat transaccions per un amount superior a la mitjana de totes les transaccions.

```
52 •  SELECT c.id,
53            c.company_name
54      FROM company c
55     WHERE EXISTS (SELECT 1
56                     FROM transaction t
57                    WHERE t.company_id = c.id
58                          AND amount > (
59                                         SELECT AVG(t.amount) AS media_total_transactions
60                                           FROM transaction t
61                                        )
62                   );
63
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

id	company_name
b-2222	Ac Fermentum Incorporated
b-2226	Magna A Neque Industries
b-2230	Fusce Corp.
b-2234	Convallis In Incorporated
b-2238	Ante Iaculis Nec Foundation

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
8	22:28:02	SELECT c.id, c.company_name FROM company c WHERE EXISTS (SELECT 1 FROM transaction t WHERE t.company_id...	100 row(s) returned	0.125 sec / 0.000 sec

Eliminaran del sistema les empreses que no tenen transaccions registrades, entrega el llistat d'aquestes empreses.

```
65 •  SELECT c.id AS company_no_transactions
66      FROM company c
67     WHERE NOT EXISTS (
68                        SELECT 1
69                          FROM transaction t
70                         WHERE c.id=t.company_id
71                       );
72
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_no_transactions

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Or

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	22:28:02	SELECT c.id AS company_no_transactions FROM company c WHERE NOT EXISTS (SELECT 1 FROM transaction t WHERE c.id=t.compan...	0 row(s) returned	0.016 sec / 0.000 sec

Nivell 2

Exercici 1

Identifica els cinc dies que es va generar la quantitat més gran d'ingressos a l'empresa per vendes. Mostra la data de cada transacció juntament amb el total de les vendes.

```
SELECT DATE(timestamp) AS day,
       SUM(amount) AS total_amount
  FROM transaction
 GROUP BY DATE(timestamp)
 ORDER BY total_amount DESC
 LIMIT 5;
```

day	total_amount
2022-12-13	14337.44
2019-11-18	13591.32
2023-02-20	13332.59
2017-12-20	13318.43
2019-03-18	12680.95

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14

Output

Action Output

#	Time	Action	Message	Duration / Fetch
10	22:28:02	SELECT DATE(timestamp) AS day, SUM(amount) AS total_amount FROM transaction GROUP BY DATE(timestamp) ORDER BY total_amount D...	5 row(s) returned	0.140 sec / 0.000 sec

Exercici 2

Quina és la mitjana de vendes per país? Presenta els resultats ordenats de major a menor mitjà.

```
SELECT c.country,
       ROUND(AVG(amount),2) AS avg_sales
  FROM transaction t
  JOIN company c
    ON t.company_id = c.id
 GROUP BY c.country
 ORDER BY avg_sales DESC;
```

country	avg_sales
Australia	265.19
United States	264.98
Belgium	261.15
Germany	260.84
Ireland	260.64
Spain	260.47
France	259.98
New Zealand	259.59
Norway	259.38
Netherlands	258.44
Italy	258.27
Sweden	257.44
Canada	257.43
United Kingdom	256.76
China	252.42

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14

Output

Action Output

#	Time	Action	Message	Duration / Fetch
11	22:28:03	SELECT c.country, ROUND(AVG(amount),2) AS avg_sales FROM transaction t JOIN company c ON t.company_id = c.id GROUP BY c.coun...	15 row(s) returned	0.375 sec / 0.000 sec

Exercici 3

En la teva empresa, es planteja un nou projecte per a llançar algunes campanyes publicitàries per a fer competència a la companyia "Non Institute". Per a això, et demanen la llista de totes les transaccions realitzades per empreses que estan situades en el mateix país que aquesta companyia.

Mostra el llistat aplicant JOIN i subconsultes.

```
SELECT t.id AS transaction_id_competitors,
       c.company_name AS competitors
  FROM transaction t
  JOIN company c
    ON t.company_id = c.id
 WHERE country = (
                  SELECT country
                    FROM company
                   WHERE company_name = "Non Institute"
                  );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

transaction_id_competitors	competitors
008629B4-C9A9-406C-A3D2-71FDA47BC546	Sed Nunc Ltd
00B72BA4-54A3-4B8E-B13F-2D57535AA17A	Sed Nunc Ltd
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	Sed Nunc Ltd
023FFCE8-E618-4938-BF56-C8DF80540ADD	Sed Nunc Ltd
02683BEB-EF91-4564-957B-D6F1662AB7C5	Sed Nunc Ltd
02C2F29E-CEF2-4C1E-A594-F476E8F279C0	Sed Nunc Ltd
02F468DC-426C-47C2-8B0A-D8B25B7A81AF	Sed Nunc Ltd
03068E3B-817B-4A49-934E-0E439291A104	Sed Nunc Ltd
0347BFE6-8EB5-4387-B187-0E78E8F2B8FB	Sed Nunc Ltd
03AEBD0E-DC97-4BD3-9C57-6A6DB78026FD	Sed Nunc Ltd

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
12	22:28:03	SELECT t.id AS transaction_id_competitors, c.company_name AS competitors FROM transaction t JOIN company c ON t.company_id = c.id W...	2000 row(s) returned	0.015 sec / 0.000 sec

Mostra el llistat aplicant solament subconsultes.

```
SELECT t.id AS transaction_id_competitors,
       t.company_id,
       (SELECT c.company_name
          FROM company c
         WHERE c.id=t.company_id) AS company
  FROM transaction t
 WHERE EXISTS (
                SELECT 1
                  FROM company c
                 WHERE c.id = t.company_id
                       AND country = (
                                      SELECT c.country
                                        FROM company c
                                       WHERE company_name = "Non Institute"
                                      )
               );
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

transaction_id_competitors	company_id	company
008629B4-C9A9-406C-A3D2-71FDA47BC546	b-2246	Sed Nunc Ltd
00B72BA4-54A3-4B8E-B13F-2D57535AA17A	b-2246	Sed Nunc Ltd
01F075B1-D7AE-4D02-AAD9-5FFD72A43F3C	b-2246	Sed Nunc Ltd
023FFCE8-E618-4938-BF56-C8DF80540ADD	b-2246	Sed Nunc Ltd
02683BEB-EF91-4564-957B-D6F1662AB7C5	b-2246	Sed Nunc Ltd
02C2F29E-CEF2-4C1E-A594-F476E8F279C0	b-2246	Sed Nunc Ltd

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	22:28:03	SELECT t.id AS transaction_id_competitors, t.company_id, (SELECT c.company_name FROM company c WHERE c.id=t.company_id) A...	2000 row(s) returned	0.000 sec / 0.015 sec

Nivell 3

Exercici 1

Presenta el nom, telèfon, país, data i amount, d'aquelles empreses que van realitzar transaccions amb un valor comprès entre 350 i 400 euros i en alguna d'aquestes dates: 29 d'abril del 2015, 20 de juliol del 2018 i 13 de març del 2024. Ordena els resultats de major a menor quantitat.

```sql
SELECT c.company_name,
       c.phone,
       DATE(t.timestamp) AS date,
       t.amount
  FROM company c
  JOIN transaction t
    ON t.company_id = c.id
 WHERE t.amount BETWEEN 350 AND 400
       AND DATE(t.timestamp) IN ('2015-04-29', '2018-07-20', '2024-03-13')
 ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_name	phone	date	amount
Aliquam PC	01 45 73 52 16	2024-03-13	399.84
Auctor Mauris Vel LLP	08 09 28 74 14	2018-07-20	399.51
At Pede Corp.	06 14 48 33 15	2015-04-29	390.69
Aliquam PC	01 45 73 52 16	2024-03-13	388.29
Orci Adipiscing Limited	03 18 00 77 81	2018-07-20	373.71
Fringilla LLC	08 29 15 93 57	2015-04-29	367.62
Pede Cum Ltd	07 62 26 48 38	2018-07-20	356.87
Auctor Mauris Vel LLP	08 09 28 74 14	2024-03-13	353.75

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read O

Output

Action Output

#	Time	Action	Message	Duration / Fetch
14	22:28:03	SELECT c.company_name, c.phone, DATE(t.timestamp) AS date, t.amount FROM company c JOIN transaction t ON t.company_id ...	8 row(s) returned	0.078 sec / 0.000 sec

Exercici 2

Necessitem optimitzar l'assignació dels recursos i dependrà de la capacitat operativa que es requereixi, per la qual cosa et demanen la informació sobre la quantitat de transaccions que realitzen les empreses, però el departament de recursos humans és exigent i vol un llistat de les empreses on especifiquis si tenen més de 400 transaccions o menys.

```sql
SELECT t.company_id,
       c.company_name,
       COUNT(t.id) AS total_transactions,
       CASE
           WHEN COUNT(t.id) >= 400 THEN 'YES'
           WHEN COUNT(t.id) < 400 THEN 'NO'
       END AS 'More than 400'
  FROM transaction t
  JOIN company c
    ON c.id=t.company_id
 GROUP BY t.company_id
 ORDER BY total_transactions;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

company_id	company_name	total_transactions	More than 400
b-2362	Lorem Eu Incorporated	380	NO
b-2286	Fringilla LLC	397	NO
b-2470	Nec Luctus LLC	399	NO
b-2578	Dui Quis Institute	402	YES
b-2446	Risus Associates	407	YES
b-2226	Magna A Neque Industries	410	YES
b-2526	Amet Luctus Vulputate Foundation	411	YES
b-2298	Elit Etiam Laoreet Associates	418	YES

Result 1 | Result 2 | Result 3 | Result 4 | Result 5 | Result 6 | company 7 | company 8 | Result 9 | Result 10 | Result 11 | Result 12 | Result 13 | Result 14 | Read Onl

Output

Action Output

#	Time	Action	Message	Duration / Fetch
15	22:28:03	SELECT t.company_id, c.company_name, COUNT(t.id) AS total_transactions, CASE WHEN COUNT(t.id) >= 400 THEN 'YES' WHEN COU...	100 row(s) returned	0.172 sec / 0.000 sec